**Alamein Petroleum Company**
**Cairo, Egypt**

## Horus-18
## A/R “G” Dolomite

# Acid Frac
# Post Job Report

For: Alamein Engineers
Date: Tuesday, October 19, 2021



HALLIBURTON

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# 1.0 Well Information

## 1.1 Customer Information

| | |
|---|---|
| **Customer** | Alamein Petroleum Company |
| **Well Name** | Horus-18 |
| **Platform/Rig Name** | Target1 |
| **Rig Type** | Workover Rig |
| **Interval** | A/R "G" Dolomite |
| **Job Date** | 19-Oct-2021 |
| **Country** | Egypt |
| **Customer Representative** | Alamein Engineers |
| **Halliburton Representative** | Mohamed Seada/Omar Khaled |

## 1.2 Pipe Information

| Equipment | Top MD m | Bottom MD m | OD in | ID in | Weight lb/ft |
|---|---|---|---|---|---|
| Surface Pipe | 0.00 | 91.44 | 4.000 | 3.000 | |
| Tubing | 0.00 | 809.63 | 3.500 | 2.750 | 12.95 |
| Tubing | 809.63 | 814.72 | 3.500 | 2.440 | |
| Tubing | 814.72 | 820.81 | 3.500 | 2.280 | |
| Tubing | 820.81 | 821.20 | 3.500 | 2.440 | |
| Tubing | 821.20 | 1766.78 | 3.500 | 2.750 | 12.95 |
| Tubing | 1766.78 | 1767.14 | 3.500 | 2.440 | |
| Tubing | 1767.14 | 1767.74 | 3.500 | 2.250 | |
| Tubing | 1767.74 | 1768.73 | 3.500 | 2.440 | |
| Packer | 1768.73 | 1771.65 | | 2.370 | |
| Tubing | 1771.65 | 1772.07 | 3.500 | 2.440 | |
| Tubing | 1772.07 | 1772.42 | 3.500 | 2.440 | |
| Tubing | 1772.42 | 1781.86 | 3.500 | 2.750 | 12.95 |
| Formation | 1812.34 | 1815.39 | | | |
| Casing | 0.00 | 1920.24 | 7.000 | 6.184 | 29.00 |

## 1.3 Perforation Intervals

| Top MD m | Bottom MD m | Number of Shots | Perf Density spf | Perf Diameter in | Perf Formation |
|---|---|---|---|---|---|
| 1812.34 | 1815.39 | 51 | 5.0 | 0.420 | PERFS |

# 2.0 Pumping Schedule

## 2.1 Designed Pumping Schedule

| Stage Number | Description | Fluid System | Clean Volume gal | Slurry Volume gal | Rate Stage Start bpm | Rate Stage End bpm | Stage Time min |
|---|---|---|---|---|---|---|---|
| 1 | Step Rate Test | 4% KCl Water | 5000 | 5000 | 25.0 | 25.0 | 4.76 |
| 2 | Shut-In | | 0 | 0 | 0.0 | 0.0 | 0.00 |
| 3 | Acid | CSA 20%-Gelled Acid | 1000 | 1000 | 10.0 | 10.0 | 2.38 |
| 4 | Pump-In | DELTA FRAC 200 30# | 2000 | 2000 | 20.0 | 20.0 | 2.38 |
| 5 | Acid | CEA 20% | 2250 | 2250 | 20.0 | 20.0 | 2.68 |
| 6 | Pump-In | DELTA FRAC 200 30# | 1000 | 1000 | 20.0 | 20.0 | 1.19 |
| 7 | Acid | CSA 20%-Gelled Acid | 4500 | 4500 | 22.0 | 22.0 | 4.87 |
| 8 | Pump-In | AGS | 4500 | 4500 | 22.0 | 22.0 | 4.87 |
| 9 | Pump-In | DELTA FRAC 200 30# | 3000 | 3000 | 20.0 | 20.0 | 3.57 |
| 10 | Acid | CEA 20% | 3250 | 3250 | 20.0 | 20.0 | 3.87 |
| 11 | Pump-In | DELTA FRAC 200 30# | 1500 | 1500 | 20.0 | 20.0 | 1.79 |
| 12 | Acid | CSA 20%-Gelled Acid | 7000 | 7000 | 22.0 | 22.0 | 7.58 |
| 13 | Shut-In | | 0 | 0 | 0.0 | 0.0 | 0.00 |
| 14 | Acid | Tank Bottoms | 3000 | 3000 | 5.0 | 5.0 | 14.29 |
| 15 | Flush | 4% KCl Water | 8000 | 8000 | 10.0 | 10.0 | 19.05 |
| 16 | Shut-In | | 0 | 0 | 0.0 | 0.0 | 0.00 |
| Total | | | 46000 | 46000 | | | 73.27 |

# 3.0 Actual Stage Summary

## 3.1 Stage Summary

| Stage Number | Start Time ucts | Max Treat Pr psi | Avg Treating Pressure psi | Max Slurry Rate bpm | Avg Slurry Rate bpm | Avg Clean Rate bpm | Slurry Volume gal | Clean Volume gal | Avg Hydraulic Horsepower hp |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 09:55:27 | 8500 | 5011 | 22.5 | 15.1 | 15.1 | 6462 | 6462 | 1859 |
| 2 | 10:06:14 | 0 | 0 | 0.0 | 0.0 | 0.0 | 0 | 0 | 0 |
| 3 | 10:47:03 | 56 | 21 | 13.2 | 11.9 | 11.9 | 990 | 990 | 6 |
| 4 | 10:49:14 | 3296 | 2348 | 19.4 | 11.2 | 11.2 | 2057 | 2057 | 647 |
| 5 | 10:53:52 | 7775 | 4923 | 20.1 | 19.6 | 19.6 | 2291 | 2291 | 2366 |
| 6 | 10:56:39 | 7647 | 7103 | 20.3 | 19.1 | 19.1 | 1001 | 1001 | 3323 |
| 7 | 10:57:54 | 6817 | 6126 | 24.9 | 23.5 | 23.5 | 4564 | 4564 | 3521 |
| 8 | 11:02:32 | 7811 | 6981 | 25.2 | 22.8 | 22.8 | 4577 | 4577 | 3894 |
| 9 | 11:07:20 | 7114 | 4542 | 21.5 | 21.2 | 21.2 | 3006 | 3006 | 2355 |
| 10 | 11:10:43 | 8232 | 5375 | 21.5 | 20.2 | 20.2 | 3365 | 3365 | 2659 |
| 11 | 11:14:41 | 7553 | 6755 | 23.7 | 21.1 | 21.1 | 1515 | 1515 | 3485 |
| 12 | 11:16:23 | 6102 | 5189 | 25.3 | 24.2 | 24.2 | 6445 | 6445 | 3080 |
| 13 | 11:22:44 | 1033 | 747 | 9.4 | 6.5 | 6.5 | 32 | 32 | 120 |
| 14 | 11:23:04 | 505 | 110 | 5.2 | 4.3 | 4.3 | 1361 | 1361 | 12 |
| 15 | 11:31:07 | 29 | 22 | 10.9 | 7.0 | 7.0 | 7994 | 7994 | 4 |
| 16 | 11:58:19 | 27 | 27 | 7.5 | 7.5 | 7.5 | 11 | 11 | 5 |

# 4.0 Performance Highlights

## 4.1 Step Rate Test Job Summary

| | | |
|---|---|---|
| Start Time | 19-Oct-21 09:11:39 | ucts |
| End Time | 19-Oct-21 10:59:30 | ucts |
| Pump Time | 10.17 | min |
| Start Averaging Time | 19-Oct-21 09:55:27 | ucts |
| End Averaging Time | 19-Oct-21 10:59:30 | ucts |
| Max Treating Pressure | 8500 | psi |
| Max Slurry Rate | 22.5 | bpm |
| Avg Treating Pressure | 5011 | psi |
| Avg Slurry Rate | 15.1 | bpm |
| Clean Volume | 6462 | gal |
| Slurry Volume | 6462 | gal |

## 4.2 Main Frac Job Summary

| | | |
|---|---|---|
| Start Time | 19-Oct-21 09:11:39 | ucts |
| End Time | 20-Oct-21 12:07:38 | ucts |
| Pump Time | 80.00 | min |
| Max Treating Pressure | 8231 | psi |
| Avg Treating Pressure | 5780 | psi |
| Max Slurry Rate | 25.3 | bpm |
| Avg Slurry Rate | 22.0 | bpm |
| Slurry Volume | 45670 | gal |

## 4.3 Volumes Summary

| Volumes Pumped | Total | Units |
|---|---|---|
| Linear Gel 30# (Main Frac) | 2913 | gal |
| Xlinked Gel 30# (Main Frac) | 7087 | gal |
| Carbonate Stimulation Acid (CSA) | 14000 | gal |
| Carbonate Emulsion Acid (CEA) | 7000 | gal |
| Guidon AGS Service Treatment | 5000 | gal |

## 4.4 Fluid Design

➢ ***Carbonate Stimulation Acid (CSA) – 14,000 gal***

| No# | Name | Function | Concentration | Volume |
|---|---|---|---|---|
| 1 | HCL Acid | Acid | 28 % | 9,637 gal |
| 2 | Fresh Water | Water | | 2,557 gal |
| 3 | FE-1A | Acetic Acid | 50 gal/Mgal | 700 gal |
| 4 | Fe-2 | Iron Sequestering Agent | 50 Ib/Mgal | 700 Ib |
| 5 | HAI-501 | Corrosion Inhibitor | 10 gal/Mgal | 140 gal |
| 6 | DCA-32023 | Surfactant | 5 gal/Mgal | 70 gal |
| 7 | AS-7 | Anti Sludging Agent | 10 gal/Mgal | 140 gal |
| 8 | SGA-HT | Gelling agent | 4 gal/Mgal | 56 gal |
| 9 | Musol-A | Mutual Solvent | 50 gal/Mgal | 700 gal |

➢ ***Carbonate Emulsion Acid (CEA) – 7,000 gal***

| Carbonate Emulsion Acid (CEA) | | | | |
|---|---|---|---|---|
| No# | Name | Function | Concentration | Volume |
| 1 | HCL Acid | Acid | 28 % | 3,442 gal |
| 2 | Fresh Water | Water | | 1,158 gal |
| 3 | AF-70 | Emulsifier | 10 gal/Mgal | 70 gal |
| 4 | FE-1A | Acetic Acid | 50 gal/Mgal | 250 gal |
| 5 | Fe-2 | Iron Sequestering Agent | 50 Ib/Mgal | 250 Ib |
| 6 | HAI-501 | Corrosion Inhibitor | 10 gal/Mgal | 50 gal |
| 7 | Diesel | Diesel | 290 gal/Mgal | 2,030 gal |

➢ ***Guidon AGS Service Treatment – 5,000 gal***

| No# | Name | Function | Concentration | Volume |
|---|---|---|---|---|
| 1 | HPT-1 | Diverting Agent | 67 gal/Mgal | 335 gal |
| 2 | Fresh Water | Water | 933 gal/Mgal | 4,665 gal |
| 3 | KCL | Salt | 2 Ib/Mgal | 777 Ib |

### ➢ *Flush – 8,000 gal*

| No# | Name | Function | Concentration | Volume |
|---|---|---|---|---|
| 1 | Fresh Water | Water | 949 gal/Mgal | 7,592 gal |
| 2 | Musol A | Mutual Solvent | 50 gal/Mgal | 400 gal |
| 3 | DCA-32023 | Surfactant | 2 gal/Mgal | 16 gal |
| 4 | KCL | Salt | 333 lb/Mgal | 2,528 lb |

### ➢ *Step Rate – 5,000 gal*

| No# | Name | Function | Concentration | Volume |
|---|---|---|---|---|
| 1 | Fresh Water | Water | 949 gal/Mgal | 4,745 gal |
| 2 | DCA-32023 | Surfactant | 2 gal/Mgal | 10 gal |
| 3 | KCL | Salt | 333 lb/Mgal | 1,580 lb |

Disclaimer: The average and maximum values (except volumes and bottom hole values) are based on the start and end averaging times.

# 5.0 Quality Assurance

## 5.1 Base Fluid Source

| Name |
|---|
| 4% KCL Water |
| 4% KCL Water |

## 5.2 Break Test

| Test Number | Fluid Temp °F | Fluid Name | SP breaker gal/Mgal |
|---|---|---|---|
| 1 | 160.0 | DELTA FRAC 200 30# | 1.00 |

| Test Duration min | Test #1 Fluid Visc |
|---|---|
| 10.00 | C-Crosslinked |
| 20.00 | C-Crosslinked |
| 30.00 | C-Crosslinked |
| 40.00 | C-Crosslinked |
| 50.00 | C-Crosslinked |
| 60.00 | WC-Weak crosslink |
| 70.00 | B-Broke |

## 5.3 Fluid Stability Testing

| Tank # | Fluid Name | Initial | | | After 2 Hours | | | After 4 Hours | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | Gel Loading Lbs/Mgal | Test Temp °F | Viscosity cp | Gel Loading Lbs/Mgal | Test Temp °F | Viscosity cp | Gel Loading Lbs/Mgal | Test Temp °F | Viscosity cp |
| 1 | Delta Frac 30# | 30 | 75 | 29 | 30 | 75 | 29 | 30 | 75 | 29 |
| 2 | Delta Frac 30# | 30 | 75 | 28 | 30 | 75 | 28 | 30 | 75 | 28 |

## 5.4 Crosslink Test

| Fluid Name | X-Linker | XLinker Conc gal/Mgal | Accel | Accel Conc gal/Mgal | Fluid Temp °F | Xlinker pH | Base Fluid pH | Base Fl Visc cp | Vortex min | Lip min | Clean min |
|---|---|---|---|---|---|---|---|---|---|---|---|
| DELTA FRAC 200 30# | BC-200 Crosslinker | 1.20 | CL-31 | 0.60 | 75.0 | 9.40 | 7.30 | 29.00 | 0.12 | 0.42 | 0.50 |

## 5.5 Fluid Analysis Tests

| Tank | Fluid Name | Fluid Density lb/gal | Test Temp °F | pH | Iron Conc mg/L | Phosphates Conc mg/L | Sulfate Conc mg/L |
|---|---|---|---|---|---|---|---|
| 1 | 4% KCL Water | 8.55 | 75.0 | 7.44 | 0.62 | 0.23 | 35.00 |
| 2 | 4% KCL Water | 8.55 | 75.0 | 7.26 | 0.71 | 0.16 | 44.00 |

## 5.6 Real-Time Job QA/QC

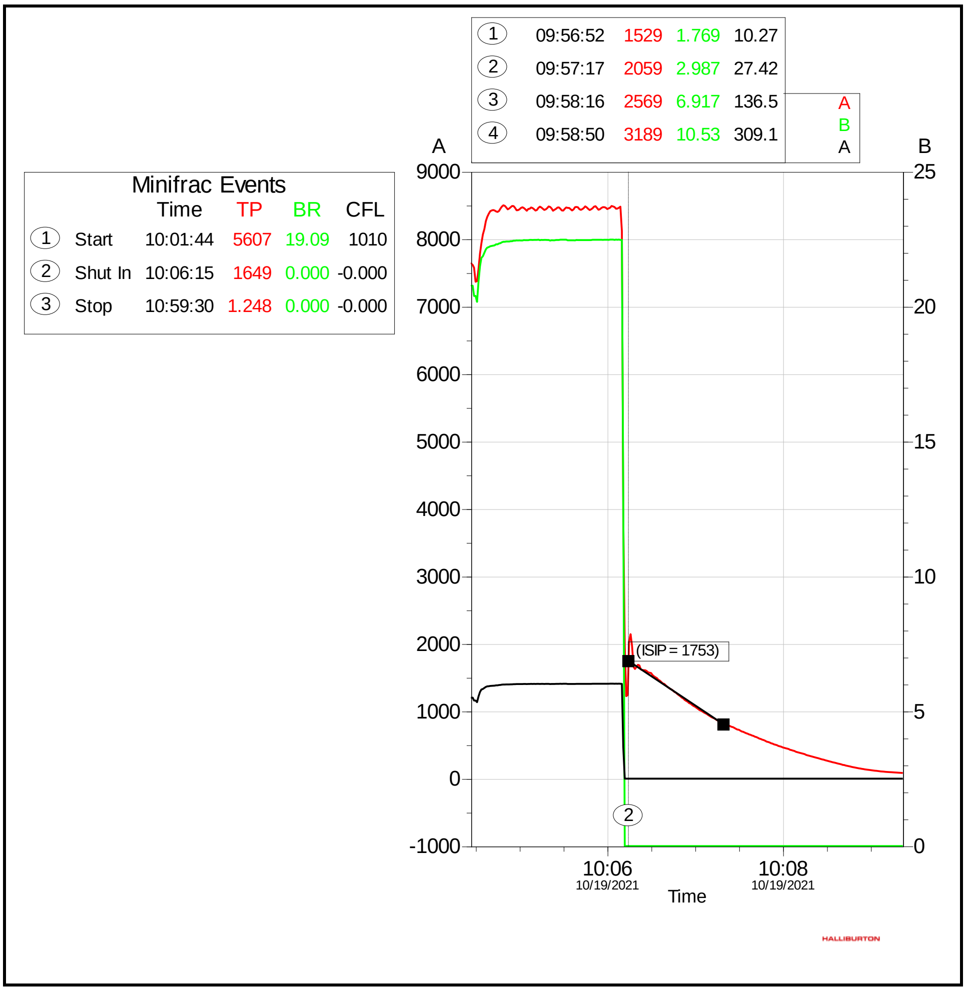
| Time (Hr:Min) | Stage Description | Fluid Type | Crosslink pH | Crosslink Time (Sec) |
|---|---|---|---|---|
| 10:49 | Delta Frac 30# | Delta Frac 30# | 9.41 | 25 |
| 10:55 | Delta Frac 30# | Delta Frac 30# | 9.39 | 26 |
| 11:06 | Delta Frac 30# | Delta Frac 30# | 9.45 | 27 |
| 11:13 | Delta Frac 30# | Delta Frac 30# | 9.4 | 25 |

# 6.0 Attachments

## 6.1 Step Rate Test Analysis

| | |
|---|---|
| Pressure Used: | Surface |
| Treating Pressure Source: | Logical - Treating Pressure (psi) |
| Treating Rate Source: | Logical - BH Rate (bpm) |
| Average Pump Rate: | 20.67 bpm |
| Pump Time: | 4.4410 min |

- ***Job Data***



## ➢ *Analysis*

| | |
|---|---|
| KANE-Pad: | 60.90 % |
| N-Pad: | 65.90 % |
| S-Pad: | 63.99 % |
| Bottom Hole ISIP: | 4475 psi |
| Fracture Gradient: | 0.75 psi/ft |
| Closure Pressure: | 3349 psi |
| Closure Gradient: | 0.56 psi/ft |
| Fluid Efficiency: | 21.96 % |
| Closure Time: | 19-Oct-2021 10:07:40 |

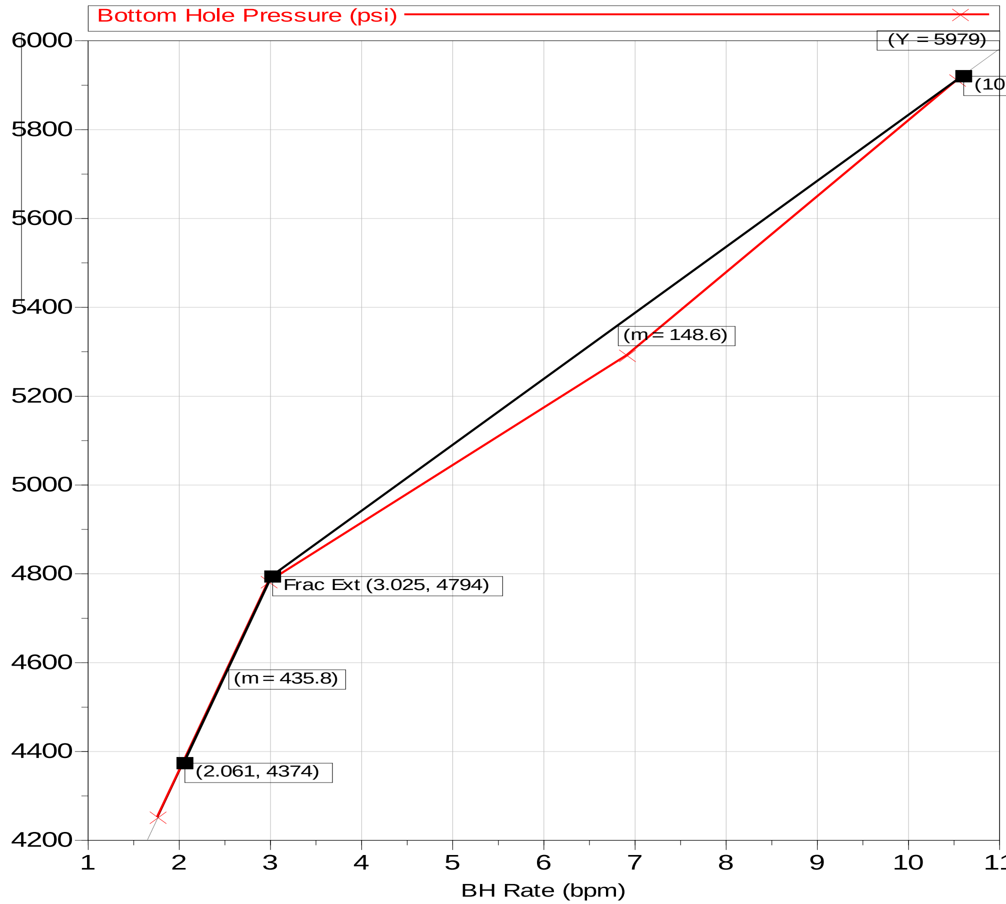
## Graph Table

| Graph | Closure Pressure (psi) | Bottom Hole ISIP (psi) | Fracture Extension Pressure (psi) | Delta Pressure (psi) | Fluid Efficiency (%) |
|---|---|---|---|---|---|
| | | 4475 | | | |
| Step Up Test | | | 4794 | | |
| Minifrac - Square Root | 3349 | | | 1126 | 21.96 |
| Minifrac - Log Log | 3349 | | | 1126 | 21.96 |
| Minifrac - G Function | 3349 | | | 1126 | 21.96 |

**Fluid efficiency is estimated to be 21.96% however the main objective was to increase rate in steps which had effect on fluid efficiency accuracy.

**3349 psi is a potential closure point that  is not confirmed due to surface pressure reaching zero.

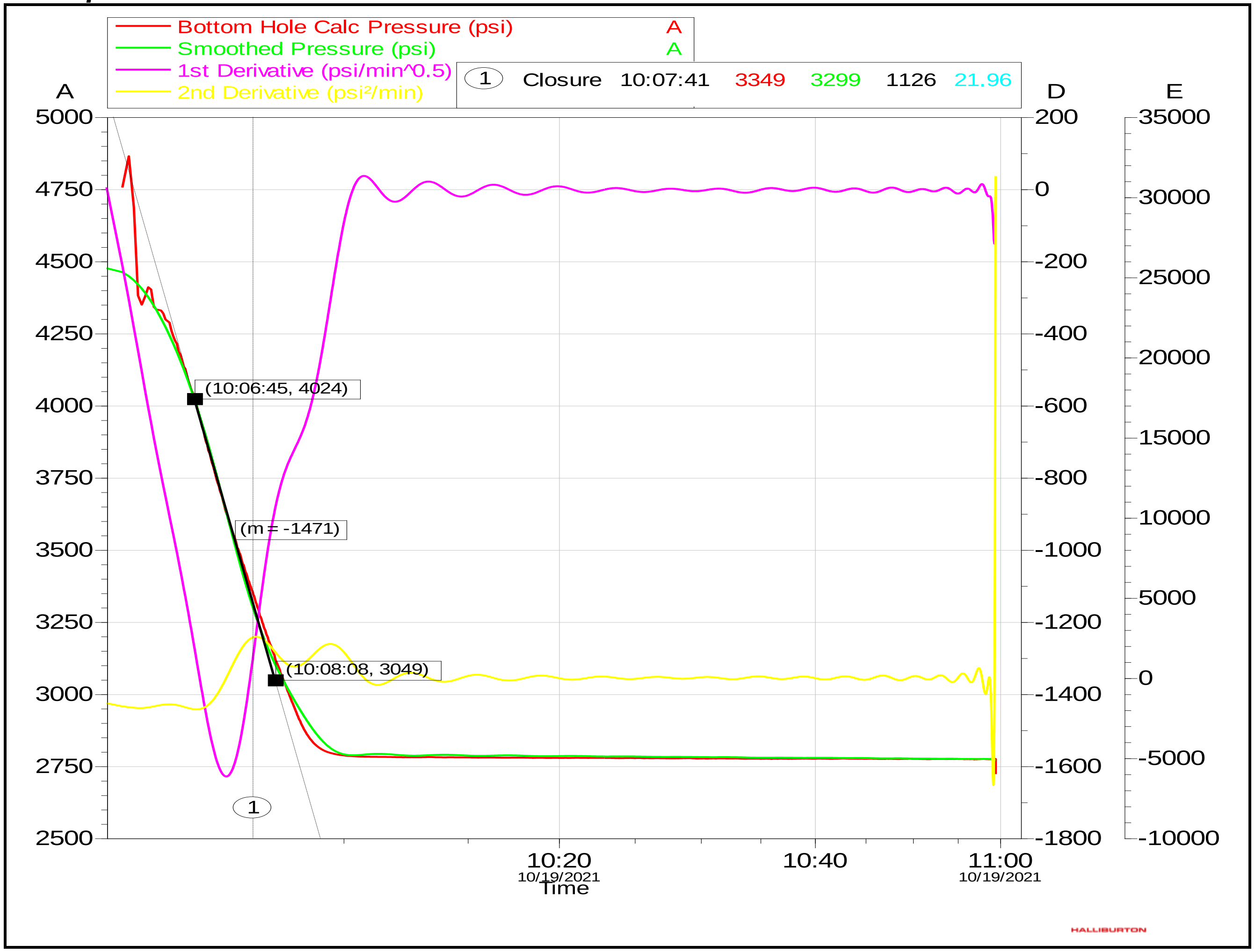
## ➢ *Step Up Test*



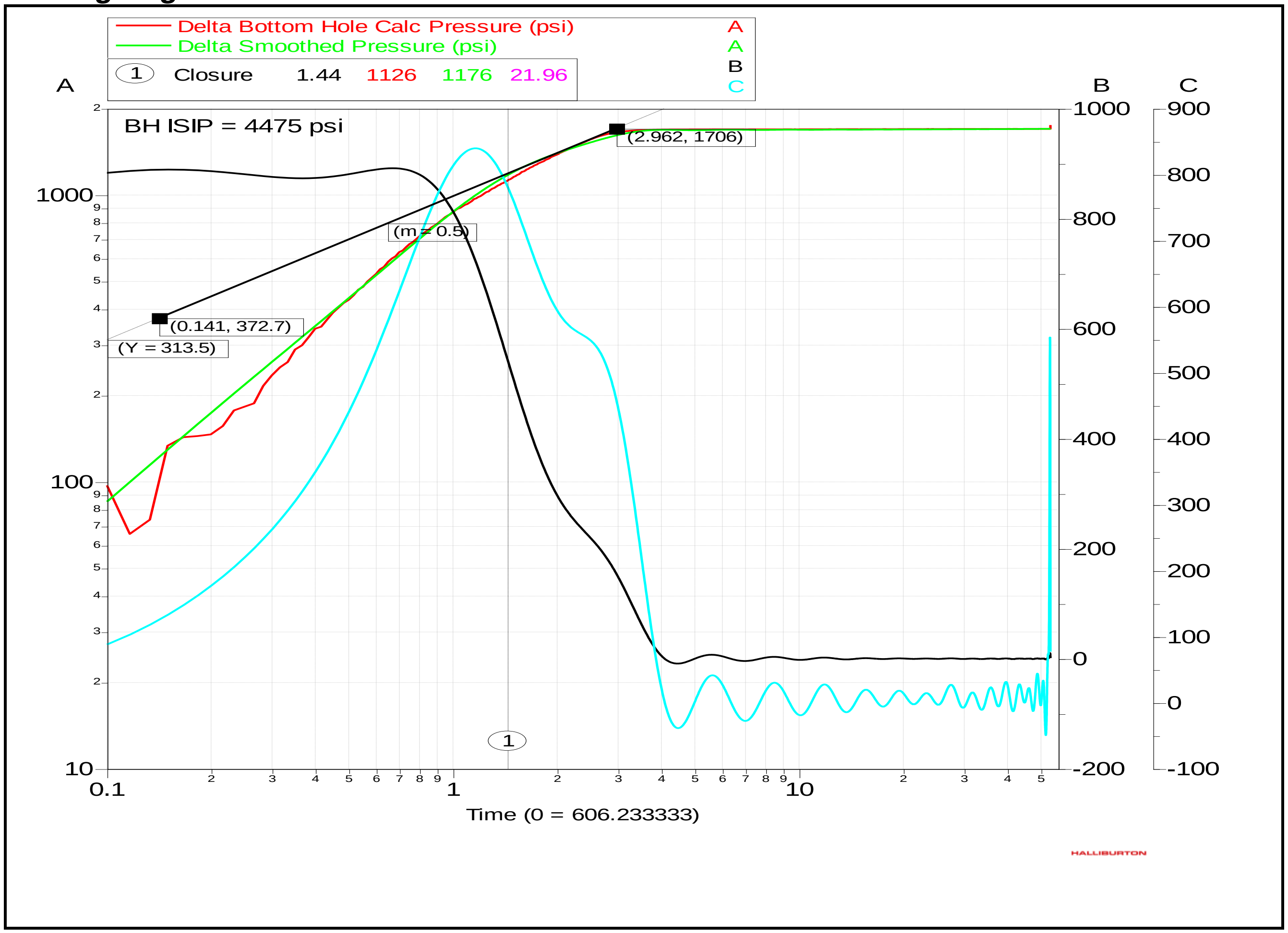

Step Rate Events

| Time | BHP | BR |
|---|---|---|
| 09:56:52 | 4251 | 1.769 |
| 09:57:17 | 4781 | 2.987 |
| 09:58:15 | 5291 | 6.917 |
| 09:58:50 | 5910 | 10.54 |

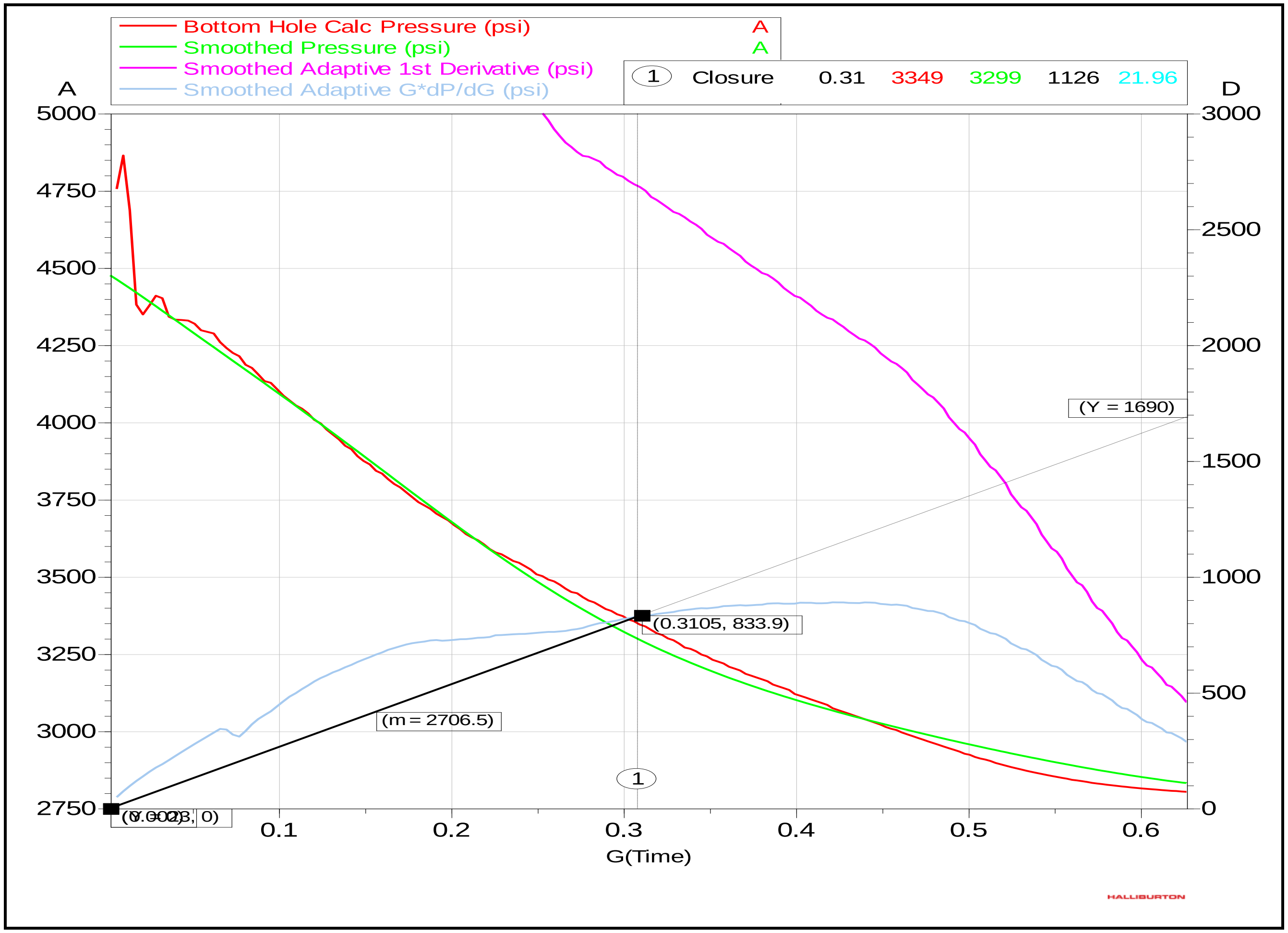
- ***Minifrac - Square Root***



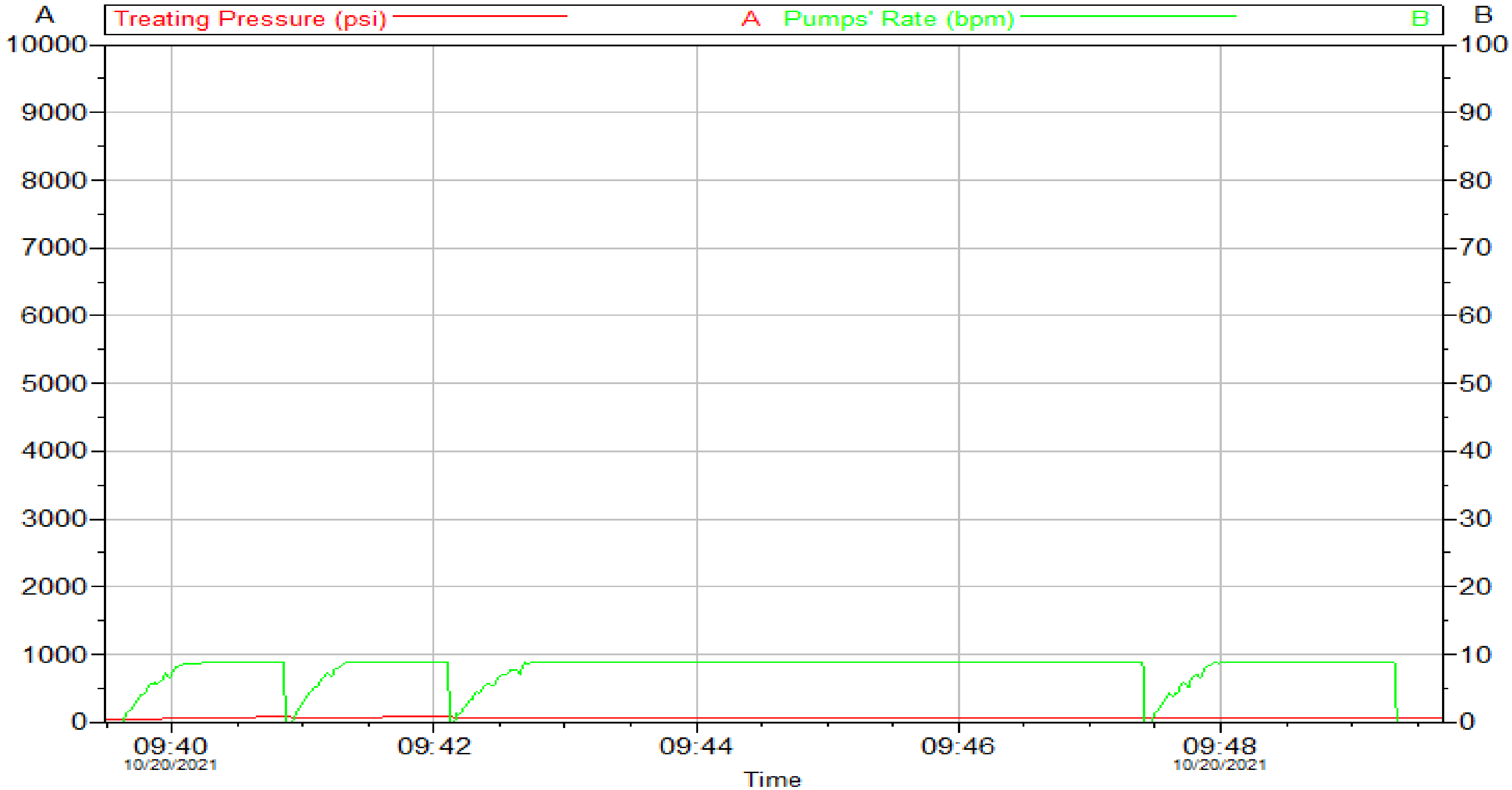
➢ ***Minifrac - Log Log***



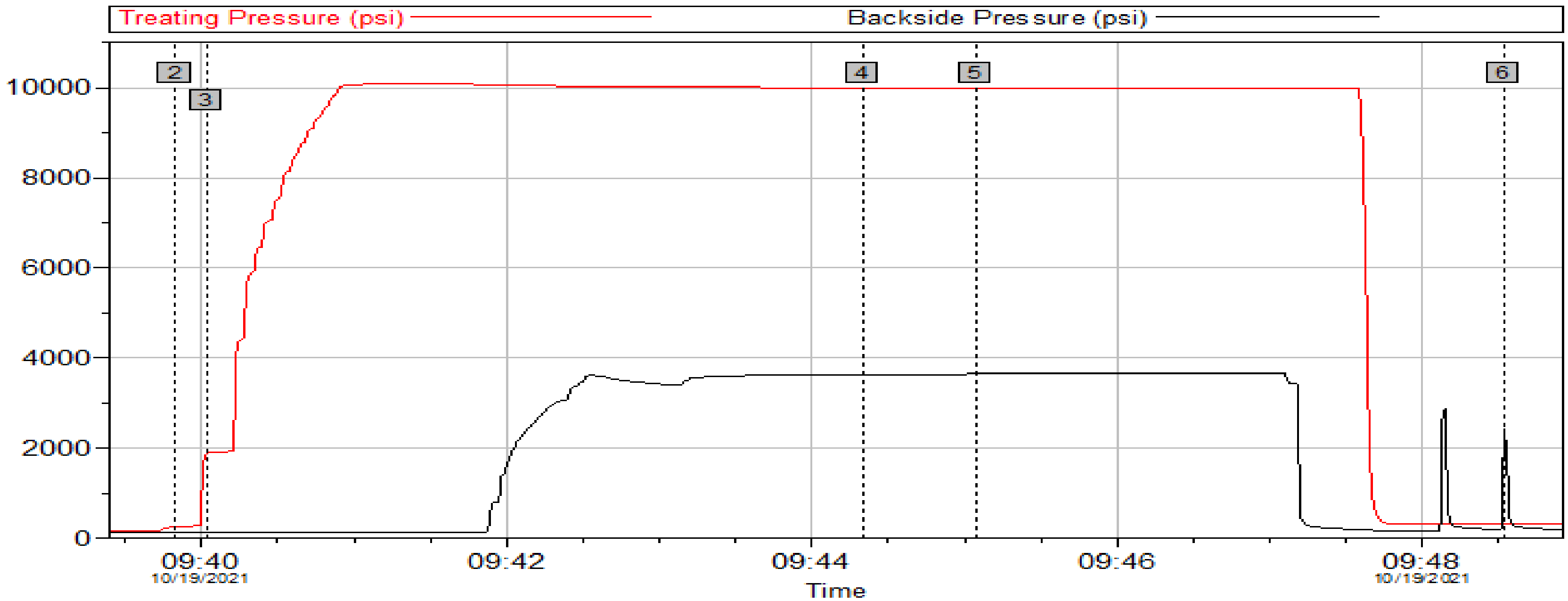
## ➢ *Minifrac - G Function*



## 6.2 High Pressure Pumps Priming



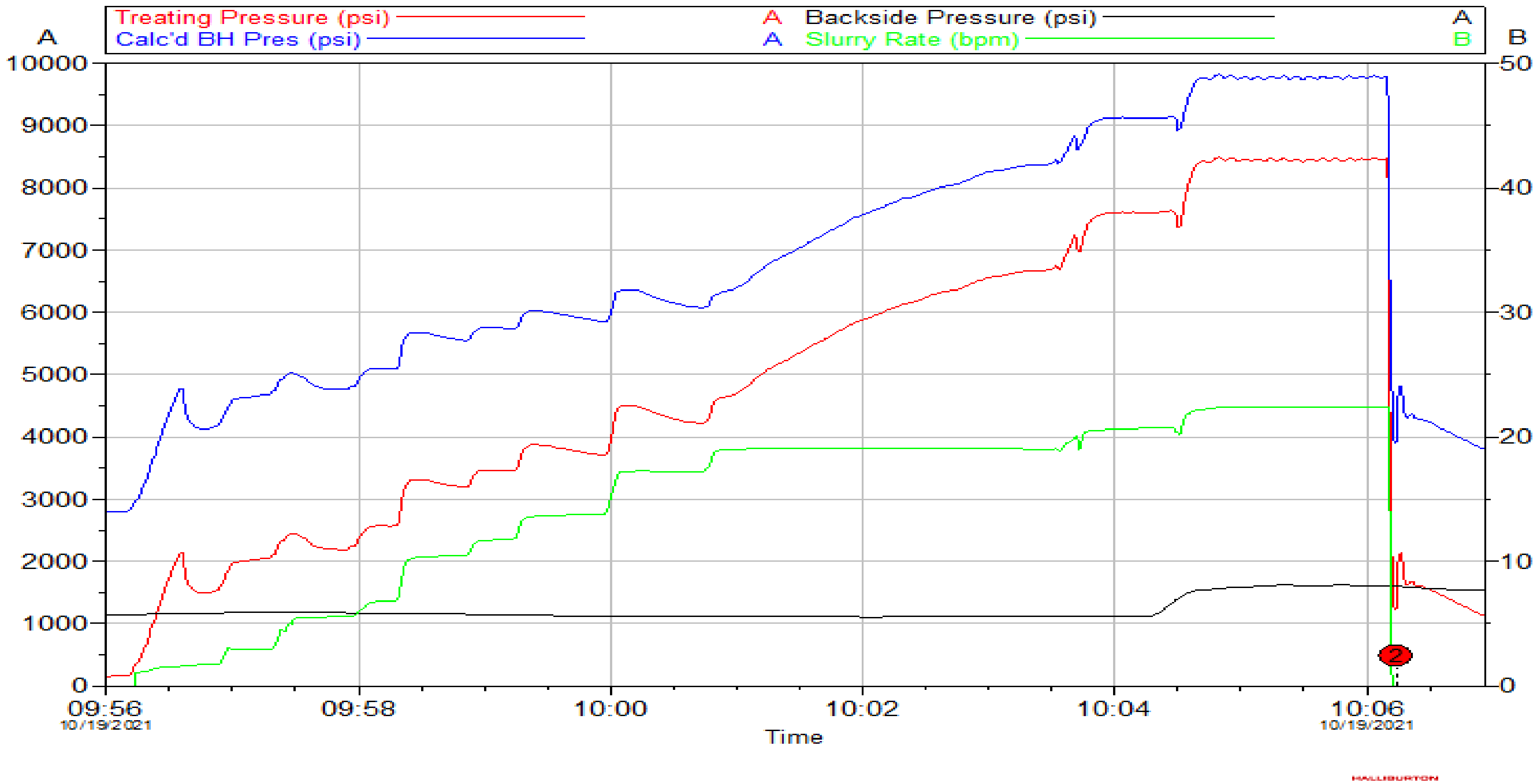
## 6.3 Pressure Test Summary




Pressure Test Summary

| | Intersection | Treating Pr | Backside Pr | | Intersection | Treating Pr | Backside Pr |
|---|---|---|---|---|---|---|---|
| 2 | Global Kickout System | 251.6 | 146.0 | 3 | Local Kickout System | 1871 | 145.9 |
| 4 | Treating Lines Pressure Test | 9999 | 3640 | 5 | Backside Line Pressure Test | 9993 | 3646 |
| 6 | Backside Line PRV | 307.2 | 2365 | | | | |

## 6.4 Step Rate Test




## 6.5 Main Frac Chemical Additives Job Summary

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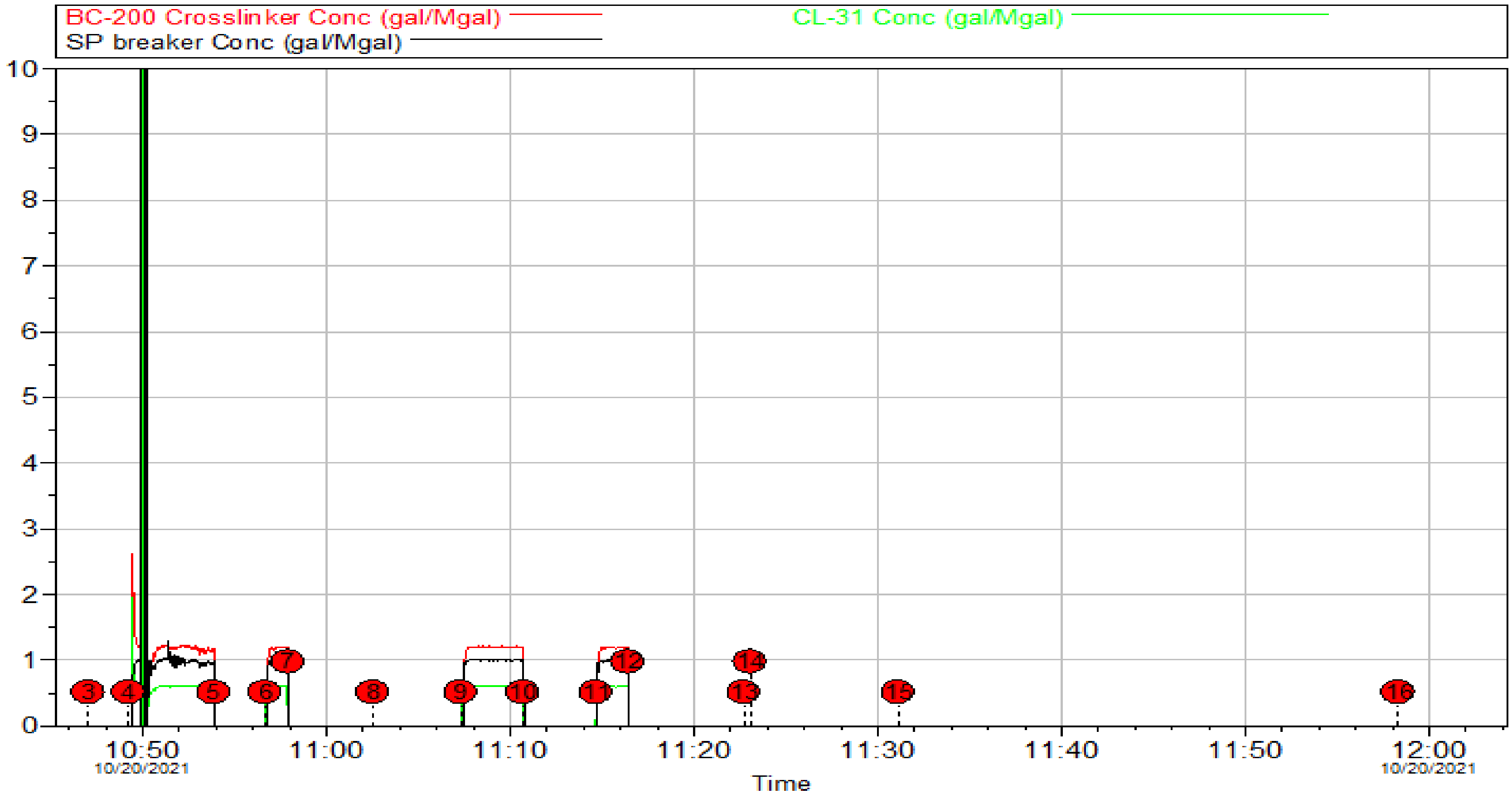

## 6.6 Acid Frac Job Summary